Curriculum Vitae

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Profile

Independent researcher in number theory with a focus on Diophantine equations and explicit constructions. Author of multiple preprints addressing classical problems such as the sum of three cubes, taxicab numbers, and the perfect cuboid problem, with work disseminated via ArXiv and Figshare. Contributor of two accepted integer sequences to the OEIS. Actively engaged with the mathematical community through MathOverflow discussions and correspondence with established researchers. Research interests lie at the intersection of symbolic methods, computational experimentation, and classical arithmetic questions.

Education

Bachelor of Science in Physician Assistantship 2020–2021
University for Development Studies, Tamale, Ghana.
Postgraduate Taaljaar Nederlands voor Anderstaligen 2021–2022
Linguapolis

Following language studies, briefly studied bioengineering (2022-2024) and computer science (2024-2025).

Research and Publications

Preprints and Articles (2025)

- **A Parametric Framework for the Sum of Three Cubes Problem** *Nov 2025 (In progress)*
  Figshare
  Developing novel approaches to the sum of three cubes problem
- **Generating Infinitely Many Taxicab Numbers** *Oct 2025 (In progress)*
  Figshare
  Constructive methods for generating taxicab numbers
- **Advancing the 3×3 Magic Square of Squares** *Sep 2025*
  Figshare
  New insights into a combinatorial number theory problem
- **Divisor-Based Impossibility Constraints for Perfect Cuboids** *Oct 2025*
  Figshare
  Improved constraints on perfect cuboid existence
- **On Equal Sums of Fourth Powers** *Aug 2025*
  Figshare

Computational and symbolic approaches to $A^4 + B^4 = C^4 + D^4$

- **A Closed-Form Symbolic Generator for $A^n + B^n = C^n + D^n$** *Jun 2025*
  arXiv:2506.19173
  Harvard ADS
  Closed-form solutions for $n = 2$ and $n = 3$ cases
- **A Divisor-Based Proof on the Non-Existence of Perfect Cuboids** *Apr 2025*
  Figshare
  Early work on perfect cuboid problem

Ongoing Research Directions

- Bridging combinatorics and number theory with applications to generalized taxicab numbers
- Extending symbolic generator framework to higher powers ($n = 4$, $n = 5$)
- Formalizing perfect cuboid non-existence proof in Lean theorem prover
- Continued work on the 3×3 magic square of squares problem

Seminars Attended

1. **PIMS Colloquium: AI and Mathematics** — Terence Tao (UCLA) Sep 2024
2. **PIMS Colloquium: The question mark function, welding and complex dynamics** — Curtis McMullen (Harvard) Sep 2025
3. **PIMS Colloquium: A Conjecture of Smyth** — Jordan Ellenberg (Wisconsin) Feb 2025
4. **MoMath: Meet a Mathematician** — Ingrid Daubechies and Lilian Pierce Nov 2023
5. **Number Theory Web Seminar: Erdős's Integer Dilation Approximation** — Dimitris Koukopoulos Apr 2025
6. **Number Theory Web Seminar: Szpiro Ratios of Elliptic Curves** — Stephanie Chan Apr 2025
7. **Number Theory Web Seminar: Axiomatic multiple Dirichlet series** — Will Sawin Jun 2025

Achievements

- **OEIS Sequence A384106** — Discovered and published new integer sequence
  On-Line Encyclopedia of Integer Sequences
- **OEIS Sequence A389865** — Additional sequence accepted to OEIS
  On-Line Encyclopedia of Integer Sequences

Skills and Languages

Technical Skills

- Independent Mathematical Research
- Diophantine Equations & Number Theory
- LaTeX Document Preparation
- Computational Tools: Desmos, Wolfram Alpha
- Learning: SageMath, Lean Theorem Prover

Languages

- **English** — Fluent
- **Dutch** — Professional Working Proficiency

References and additional materials available upon request
Last updated: January 23, 2026